Ioan Assenov

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Professional Summary

Mechanical Engineering graduate from the University of Michigan with hands-on experience in structural analysis, dynamic systems, and precision instrumentation across aerospace internships, research labs, and large-scale student engineering organizations. Demonstrated ability to lead cross-functional teams, manage multi-subsystem hardware programs, and execute from design through physical validation in high-stakes environments.

Education

University of Michigan — Ann Arbor, MI

Bachelor of Science in Mechanical Engineering GPA: 3.97/4.00 Summa Cum Laude — 2025

Dean's List and University honors (7 consecutive semesters), James B. Angell Scholar.

Relevant Coursework: Autonomous Vehicles Controls and Perception, Advanced Dynamics and Vibrations, Solid Mechanics, Fluid Dynamics, Thermodynamics

Experience

Research Assistant, Vibrations and Acoustics Laboratory (PI: Dr. Kenn Oldham) — 05/2025 - Present

University of Michigan — *Ann Arbor, MI*

- Built calibration testbed for endomicroscopic laser devices using 2-axis motorized piezoelectric stage with a parametrized raster scanning script in MATLAB, enabling development of closed-loop piezo MEMS scanning mirrors for microendoscopes
- Programmed real-time data acquisition and image reconstruction algorithm in MATLAB with data validation, correction, and downsampling, generating high fidelity optical target images for resolution testing

Teaching Assistant, MechEng 360: Modeling, Analysis, and Control of Dynamics Systems — 01/2025 - 05/2025

University of Michigan — *Ann Arbor, MI*

- Collaborated closely with faculty to develop course content, homework, and exam materials, aligned with learning objectives of the University of Michigan's Mechanical Engineering program
- Hosted semiweekly office hours covering differential equations, dynamic system modeling, and vibrations for over 170 undergraduate students

Structural Engineering Intern — 07/2024 - 09/2024

DRONAMICS — *Sofia, Bulgaria*

- Devised design for new main landing gear (MLG) to be made of carbon fiber composite through static analysis, resulting in 60% weight reduction per landing gear from current design
- Conducted 4 multi-DOF dynamic analyses on aircraft MLG and determined suspension stiffness/damping parameters, ensuring safe system response during hard landing in Simulink
- Evaluated dynamics of two proposed parachute system drop tests to define test configuration parameters for 1300kg aircraft, advancing flight termination system certification progress

Systems Engineering Director — 05/2023 - 09/2024

Michigan Aeronautical Science Association (MASA, Student Organization) — *Ann Arbor, MI*

- Coordinated 7 cross-functional rocket subsystems and 50+ engineers to meet vehicle-level requirements and ensure team alignment, maintaining 8-month engine development timeline
- Developed and promoted sustainable documentation practices and standards to ensure critical knowledge across 75+ projects remains within team as members advance into industry

Projects

Limelight Rocket Engine Nozzle

Technical Project Lead

- Designed converging/diverging rocket nozzle featuring graphite throat, ablative extension, and complex internal profile for 12.5 kN kerolox rocket engine with 30 km altitude target
- Analyzed structural load cases by hand using published research and iterated on geometry; confirming results with 20+ FEA simulations in Ansys to verify survival of 25-second full thrust burn within safety factor of 2
- Machined and assembled 4 sub-components using mill and lathe, integrating with the full engine system in 4 months for May 2024 hot fire; post-mortem confirmed nozzle integrity through a hard start event

Clementine Rocket

Testing and Launch Crew

- Assembled the most powerful fully student-built kerolox rocket (5 kN) flown in an expeditious environment with 23 peer crew and successfully launched to 2.3 km in May 2023
- Operated ground support equipment in tests covering active tank pressure control systems and rocket motor plumbing during 12-week prelaunch campaign, verifying functionality of 5+ vehicle support subsystems

Skills

Technical Skills: FEA, GD&T, System Modeling and Simulation, Containerization, PCB Design, Data Processing
Software: Siemens NX, SOLIDWORKS, Onshape, Simulink, Docker, Adobe Suite, Ansys Mechanical, *nix
Programming: Python, MATLAB, C++, Machine Learning, AI Coding Tools